Alex Herman

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Software engineer building agent harnesses and a malleable OS, with a background in embedded hardware.

B.S. Computer Engineering, Cal Poly San Luis Obispo · 2026 · GPA 3.61

AGENT SYSTEMS

Agora *Creator · open-source malleable OS with an agent harness at its core*

2025 – now • **running**

`Harness Design · LLM routing`

- Created an open-source object-based operating system designed to be agent native and allow for multi-user collaboration.
- Designed the agent harness: a proactive lead agent (JEFF) handles communication, administration and ideation, while desk agents direct Codex and Claude Code workers to run many tasks at once with minimal handoffs.
- Built a plain-English scripting language that reads like a text adventure: non-programmers write automations, and in-desktop agents learn from in-world help to build features, dashboards and real changes without code (like skills but better).
- Built a declarative JavaScript painter that allows for shared modules to be used across applications and improved globally.
- Looking to adapt the design principles to a more general Agora Protocol that would allow any two devices to speak the same language and also create a distributed social media and map software that people can build together.

Conscious Claude *Senior Project · persistent autonomous agent*

2025 – 2026

`Python · Anthropic API · VPS`

- Ran 467 autonomous cycles over 30 days with working memory, persistent storage, sandboxed code execution and a web UI.
- Fixed tool-call degradation at long context with tool search, JSON summarization and compaction-based memory.
- Moved the agent to a VPS running self-modifying code under version control.

The Depths

`Feb 2026 · React · Express · Tone.js`

Interactive text-adventure game run by a Claude dungeon master. A modular game state prevents hallucination while still allowing rich description and narration. Recently adapted to a MOO architecture for online co-development.

Friendship Loom

`2026 · JavaScript · Node · Railway`

I wanted to be able to weave any friendship bracelet I could imagine, so I created a pixel art → weaving pattern solver dashboard, hosted live for free.

CoBot

`2025 · Python · Claude`

Mac/Windows app where 4th graders write VEX IQ Python with blocks or plain English, then upload straight to the robot.

HARDWARE

Hardware Engineer *California Strawberry Commission*

2024 – 2026

`KiCad · C · Python · LoRa`

- Took an agricultural IoT tracker from prototype to production PCB as principal hardware designer: multi-rail power, smart charging, GPS, Wi-Fi, SD and LoRa.
- Rewrote firmware as a state machine and built a web interface for field data upload; debugged Rev 1 boards under the microscope. Worked with Claude in the field to decrypt CAN packets, increasing the value of our product for growers.
- Presented the system to 500+ industry professionals at National Strawberry Field Day.

OTO Smart Motor

`2024 – 2025 · capstone team lead · RP2040 · C`

Designed the PCB and closed-loop PID with auto-tuning, split across both cores so control and I2C run concurrently.

Foot Mouse for Sally

`2024 · Empower Club · Arduino · SPI`

Assistive foot mouse for a user who can't type by hand; led hardware. Honorable mention, TOM Global Maker Challenge.

TEACHING

`2021 – now` CS Tutor, CSSE Tutoring Center — CS fundamentals and AI-assisted workflows
TA, Field Botany · Ocean Lifeguard & Junior Guard Instructor, City of Santa Cruz

SKILLS

languages	JavaScript, Python, C, SystemVerilog, RISC-V assembly · English, Spanish (conversational)
agents	Anthropic API, Claude Code, Codex, tool schemas, context management and compaction
tools	Node, React, Express, Linux, Git, KiCad, LTspice